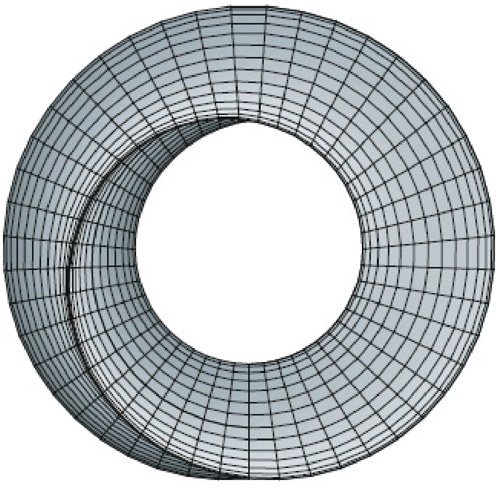
University of Zimbabwe

HMTHCS111 CALCULUS 2

Calculus of Several Variables

Author:
G. Nhawu

Department:
Mathematics and
Computational Sciences



August 28, 2026

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Chapter 4

Coordinate Geometry in 3 Dimensions

Every heart sings a song, incomplete, until another heart whispers back. Those who wish to sing always find a song. At the touch of a lover, everyone becomes a poet.

— Plato

A coordinate system represents a point in the plane by an ordered pair of numbers called coordinates. Usually we use Cartesian coordinates, which are directed distances from two perpendicular axes. Here we describe a coordinate system introduced by Newton, called the **polar coordinate system**, which is more convenient for many purposes.

So far, we have been representing graphs as collections of points (x, y) in the rectangular coordinate system.

4.1 Polar Coordinates

We fix a point O, called the **pole** (or **origin**) and construct from O an initial ray called the **polar axis**. Then each point P in the plane can be assigned **polar coordinates** (r, θ) as follows, $r =$ directed distance from O to P and $\theta =$ directed angle, counter-clockwise from polar axis to segment $\overline{OP}$.

4.1.1 Coordinate Conversion

To establish the relationship between polar and rectangular coordinates, let the polar axis coincide with the positive $x-$axis and the pole with the origin.Because (x, y) lies on a circle of radius r, it

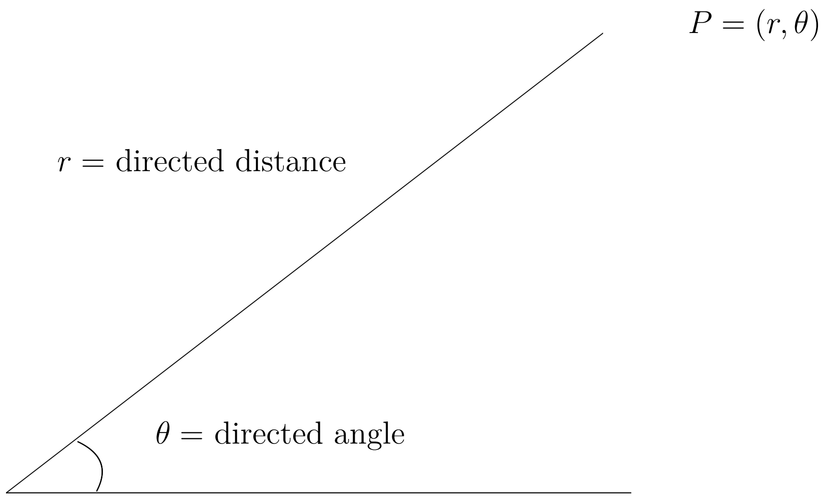

Figure 4.1: Polar Coordinates

follows that $r^2 = x^2 + y^2$. Moreover, for $r > 0$,

$$\tan\theta = \frac{y}{x}, \quad \cos\theta = \frac{x}{r}, \quad \sin\theta = \frac{y}{r}.$$

Theorem 4.1.1. *The polar coordinates (r, θ) of a point are related to the rectangular coordinates (x, y) of the point as follows,*

$$x = r\cos\theta, \quad y = r\sin\theta, \quad \tan\theta = \frac{y}{x}, \quad r^2 = x^2 + y^2.$$

Example 4.1.1. *(i) For the point $(r, \theta) = (2, \pi)$.*

$$x = r\cos\theta = 2\cos\pi = -2 \quad \textit{and} \quad y = r\sin\theta = 2\sin\pi = 0.$$

Thus, the rectangular coordinates are $(x, y) = (-2, 0)$.
(ii) For the second quadrant point $(x, y) = (-1, 1)$.

$$\tan\theta = \frac{y}{x} = -1 \Longrightarrow \theta = \frac{3\pi}{4}, \quad r = \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + 1^2} = \sqrt{2}.$$

Polar coordinates are $(r, \theta) = (\sqrt{2}, \frac{3\pi}{4})$.